Supplementary material S3: PRISMA 2020 checklist

Who Leads, Who Follows? A Review of Leadership and Guidance Dynamics in Evacuation

This review is a systematic review without meta-analysis. Its synthesis is narrative and bibliometric rather than statistical, so the PRISMA 2020 items that presuppose a pooled quantitative estimate, namely effect measures, statistical synthesis and heterogeneity, quantitative reporting-bias assessment, and formal certainty rating, do not apply. Each such item is marked *Not applicable* together with the reason. Page numbers refer to the uncolored manuscript.

Table S3.1: PRISMA 2020 checklist and the location of each item.

Item	Checklist item	Location or reason
Title		
1	Identify the report as a systematic review.	Title page; the Abstract identifies the report as a systematic review in its first sentence.
Abstract		
2	See the PRISMA 2020 for Abstracts checklist.	Abstract, p. 1.
Introduction		
3	Describe the rationale for the review in the context of existing knowledge.	Section 1, pp. 1–4; Section 2, pp. 4–6.
4	Provide an explicit statement of the objectives or questions the review addresses.	Section 3.1, p. 6; the five research questions and where each is answered are set out in Table 2, p. 7.
Methods		
5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 3.2, pp. 7–10; grouping for synthesis is described in Section 3.3, pp. 10–11.
6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted, and the date when each source was last searched.	Section 3.2, pp. 7–10; the sources of both search arms and their yields are shown in Figure 1, p. 9. The primary searches were completed in August 2024 and the supplementary search was run in August 2026, as stated in Section 3.2 and in supplementary material S1.
7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 3.2, pp. 7–10; the primary-arm strings are reproduced in Appendix A of the manuscript, and the full query log for both arms is supplied as supplementary material S1.
8	Specify the methods used to decide whether a study met the inclusion criteria, including how many reviewers screened each record and whether they worked independently.	Section 3.2, p. 7. The first author conducted the search, screening and coding; the second author independently verified the screening decisions and the coded data, with disagreements resolved through discussion.
9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report and whether they worked independently.	Section 3.3, pp. 10–11; the extracted variables are listed in Table 3, p. 11.
10a	List and define all outcomes for which data were sought.	Section 3.3 and Table 3, p. 11. The review characterizes guidance strategies, leader and follower mechanisms, compliance, and evaluation measures rather than a single clinical outcome.
10b	List and define all other variables for which data were sought.	Section 3.3 and Table 3, p. 11.
11	Specify the methods used to assess risk of bias in the included studies, including details of the tool used and how many reviewers assessed each study.	Section 3.4, pp. 11–14. Two instruments were used, one adapted from the Mixed Methods Appraisal Tool 2018 and one operationalizing established verification and validation criteria for evacuation models. The appraisal covers the 250 studies that report primary results, including those contributed by the supplementary search. A subset of ten studies was scored a second time by the first author to measure intra-rater consistency. The first author performed the appraisal for every study.
12	Specify for each outcome the effect measure used in the synthesis or presentation of results.	*Not applicable.* No pooled effect measure was computed; the synthesis is narrative and bibliometric.
13a	Describe the processes used to decide which studies were eligible for each synthesis.	Section 3.3, pp. 10–11.

Item	Checklist item	Location or reason
13b	Describe any methods required to prepare the data for presentation or synthesis.	Section 3.3, pp. 10–11, covering the coding scheme and the preparation of the term co-occurrence corpus.
13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 3.3, pp. 10–11; Figures 2–4, pp. 9–12, and Figure 6, p. 22; Tables 4–6, pp. 15–27.
13d	Describe any methods used to synthesize results.	Section 3.3, pp. 10–11. Synthesis combines thematic coding with term co-occurrence mapping; no statistical pooling was performed.
13e	Describe any methods used to explore possible causes of heterogeneity.	*Not applicable.* No statistical synthesis was performed. Variation across studies is addressed narratively in Sections 4 and 5.1.
13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	*Not applicable.* No statistical synthesis was performed.
14	Describe any methods used to assess risk of bias due to missing results in a synthesis.	*Not applicable* as a quantitative assessment. The consequences of selective and incomplete reporting are addressed in Section 3.4 and Section 5.1.
15	Describe any methods used to assess certainty in the body of evidence for an outcome.	*Not applicable.* No GRADE or comparable rating was applied, since the review does not estimate a pooled effect. Reporting completeness is characterized instead in Section 3.4 and Figure 5.
Results		
16a	Describe the results of the search and selection process, from the number of records identified through to the number of studies included in the review, ideally using a flow diagram.	Figure 1, p. 9, gives the full flow for both search arms; Section 3.2, pp. 7–10.
16b	Cite studies that might appear to meet the inclusion criteria but which were excluded, and explain why.	Figure 1, p. 9, reports the exclusion counts and reasons at each stage.
17	Cite each included study and present its characteristics.	Sections 4.1–4.4, pp. 14–30; the distribution of the corpus by year and venue is shown in Figures 2 and 3, pp. 9–10; per-study appraisal characteristics are supplied as supplementary material S2.
18	Present assessments of risk of bias for each included study.	Supplementary material S2 gives the rating of every study on every criterion; Figure 5, p. 14, summarizes them.
19	For all outcomes, present for each study summary statistics and effect estimates.	*Not applicable.* No effect estimates were extracted; per-study characteristics are given in supplementary material S2.
20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section 3.4, pp. 11–14; Figure 5, p. 14.
20b	Present results of all statistical syntheses conducted.	*Not applicable.* No statistical synthesis was conducted. The narrative and bibliometric syntheses are presented in Sections 4.1–4.4, pp. 14–30.
20c	Present results of all investigations of possible causes of heterogeneity.	*Not applicable.* No statistical synthesis was conducted.
20d	Present results of all sensitivity analyses.	*Not applicable.* No statistical synthesis was conducted.
21	Present assessments of risk of bias due to missing results in each synthesis.	Section 5.1, pp. 30–31, addresses the reporting gaps and their consequences for the evidence base.
22	Present assessments of certainty in the body of evidence.	*Not applicable.* See item 15; reporting completeness is characterized in Section 3.4 and Figure 5, p. 14.
Discussion		
23a	Provide a general interpretation of the results in the context of other evidence.	Section 5, pp. 30–34.
23b	Discuss any limitations of the evidence included in the review.	Section 5.1, pp. 30–31; Section 3.4, pp. 11–14.
23c	Discuss any limitations of the review processes used.	Section 5.1, pp. 30–31.
23d	Discuss implications of the results for practice, policy and future research.	Section 5.3, pp. 32–34; Section 6, p. 34.
Other information		

Item	Checklist item	Location or reason
24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	The review was not registered.
24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	A protocol was not registered in a public register. The review followed a predefined protocol whose eligibility criteria, search strategy, screening rules, coding scheme and appraisal instruments are set out in full in Section 3, pp. 6–14.
24c	Describe and explain any amendments to information provided at registration or in the protocol.	The supplementary targeted search reported in Section 3.2 and shown as the second arm of Figure 1, p. 9, extends the original search to social and identity-related terminology. Its criteria, sources and yield are reported in the same detail as the primary arm.
25	Describe sources of financial or non-financial support for the review, and the role of the funders.	Funding, p. 37. The review received no specific grant.
26	Declare any competing interests of review authors.	Declaration of competing interests, p. 37.
27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials.	Data availability statement, p. 38. The search query log (S1), the per-study appraisal record (S2) and the PRISMA 2020 checklist (S3) are supplied as supplementary material. The same materials, together with the term co-occurrence inputs and outputs and the source of the study selection diagram, are openly available at `https://anonymous.4open.science/r/evaclead-1E35`.